Ordinal and Latent-Grade Modelling of Radiologist-Assessed Lung Nodule Malignancy

Statistical Methods III – Project Presentation

Instructor: Prof. Shyamal Krishna De

Avigyan Chakraborty (BS2516) Oishik Basu (BS2535) Samyak Kar (BS2543)

Indian Statistical Institute, Kolkata

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Introduction

- **What is a lung nodule?** A small "spot" on the lung, typically discovered during a chest CT scan.
- **The Reality:** They are extremely common, but the vast majority are completely harmless (e.g., scars from past infections).
- **The Bottleneck:** Finding the rare, dangerous ones (early-stage cancer) is like finding a needle in a haystack. Radiologists must manually evaluate thousands of harmless spots to catch the few that matter.

Problem and Approaches

- Radiologist time is limited, especially in public health systems.
- A model linking nodule characteristics to suspected malignancy could help **prioritise review**.
- **Difficulty:** radiologists disagree.
  - In LIDC-IDRI, up to 4 radiologists rate each nodule from 1 to 5.
  - On about 78% of multi-reader training nodules, readers do *not* all agree.
- Taking one summary rating per nodule ignores this disagreement.

Two approaches compared

1. **Naive proportional odds** model of the median rating.
2. **Latent-grade model** fitted by the **EM algorithm** : the true grade is unobserved and each rating is a noisy measurement of it.

Questions we try to answer

1. Which size and shape features are associated with radiologists' malignancy ratings?
2. Does the proportional odds model adequately describe that relationship?
3. Does modelling a latent true grade change the estimated effects or the predictions?
4. Could such a model support the triage of nodules for review?

Data: LIDC-IDRI

- Lung Image Database Consortium and Image Database Resource Initiative (Armato et al., 2011), hosted on TCIA.
- **Geometric features** are shape and size measures computed from the radiologists' drawn nodule outlines (e.g. volume, solidity), whereas **semantic ratings** are the radiologists' own ordinal judgements (1–5) of a nodule's appearance (e.g. spiculation, margin).
- **Unit of analysis:** the nodule.
  - Geometric features: averaged across readers.
  - Semantic ratings: **median** across readers (ordinal data; robust to one outlying reader).

	Patients	**Nodules**
Training	609	1809
Test	261	828

Table 1: Patient-level train/test split

Response

Median malignancy $\in \{1, \ldots, 5\}$; half-grades rounded up (548 nodules, 20.8%). Individual ratings retained for the latent model.

Variables Used

Model	Variable	Meaning	Support
Primary	`log_volume`	log of volume (mm^3)	$\mathbb{R}$
	`elongation`	near 1: not elongated	$[0, 1]$
	`flatness`	near 1: not flat	$[0, 1]$
	`solidity`	voxels / convex-hull voxels	$(0, 1]$
	`compactness`	$V^{2/3}/S$	$(0, \infty)$
Secondary (adds)	`spiculation`	spiky edge extensions	$\{1, 1.5, \ldots, 5\}$
	`lobulation`	rounded edge bulges	$\{1, 1.5, \ldots, 5\}$
	`margin`	edge definition (poor $\rightarrow$ sharp)	$\{1, 1.5, \ldots, 5\}$
	`texture`	internal density	$\{1, 1.5, \ldots, 5\}$
	`sphericity`	roundness (linear $\rightarrow$ round)	$\{1, 1.5, \ldots, 5\}$

The geometric features come from radiologist-drawn outlines; the semantic ratings come from the same radiologists who rate malignancy.

Summary Statistics (Training Set, $n = 1809$)

Model	Variable	Mean	Min	Max	Std. Dev.
Primary	Log Volume	5.118	2.442	10.345	1.394
	Elongation	0.753	0.306	0.988	0.121
	Flatness	0.555	0.000	0.921	0.164
	Solidity	0.795	0.228	1.000	0.106
	Compactness	0.177	0.058	0.312	0.044
Secondary	Spiculation	1.476	1.000	5.000	0.889
	Lobulation	1.572	1.000	5.000	0.883
	Margin	3.899	1.000	5.000	1.138
	Texture	4.363	1.000	5.000	1.202
	Sphericity	3.789	1.000	5.000	0.826

Correlations Between Predictors

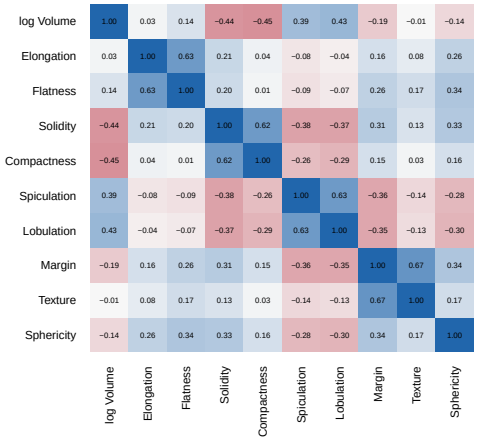


Figure 1: Correlations (training set)

- Several predictors are strongly related (e.g. shape measures with each other).
- Semantic ratings overlap with geometric shape.

Modelling an Ordinal Response

- $Y_i \in \{1, \ldots, 5\}$: median malignancy grade of nodule i; $x_i \in \mathbb{R}^p$: predictors.
- The categories are *ordered* but the gaps are not meaningful $\Rightarrow$ neither linear regression nor nominal (multinomial) logit is appropriate.
- **Cumulative probabilities** respect the ordering:

$$\gamma_{ik} = \mathbb{P}(Y_i \le k \mid x_i), \qquad 0 = \gamma_{i0} < \gamma_{i1} \le \cdots \le \gamma_{i5} = 1.$$

- Category probabilities are differences: $\mathbb{P}(Y_i = k \mid x_i) = \gamma_{ik} - \gamma_{i,k-1}$.
- **Idea:** link each cumulative probability to a linear predictor, with one intercept (threshold) per cut-point:

$$\text{logit}\ \mathbb{P}(Y_i \le k \mid x_i) = \alpha_k - x_i^\top \beta, \qquad k = 1, \ldots, 4, \quad \alpha_1 < \cdots < \alpha_4.$$

Proportional Odds: Latent-Variable Derivation

Suppose there is an unobserved continuous "malignancy suspicion" score

$$Y_i^* = x_i^\top \beta + \varepsilon_i, \qquad \varepsilon_i \sim \text{Logistic}(0,1),$$

with CDF $F(t) = \dfrac{e^t}{1+e^t}$. The observed grade is a coarsening of Y_i^*:

$$Y_i = k \iff \alpha_{k-1} < Y_i^* \le \alpha_k, \quad \alpha_0 = -\infty,\ \alpha_5 = \infty.$$

Then

$$\begin{aligned}\mathbb{P}(Y_i \le k \mid x_i) &= \mathbb{P}(\varepsilon_i \le \alpha_k - x_i^\top \beta) \\ &= F(\alpha_k - x_i^\top \beta),\end{aligned}$$

which is exactly the cumulative logit model.

Consequences

- Odds of $\{Y > k\}$ vs. $\{Y \le k\}$:

$$\frac{\mathbb{P}(Y > k \mid x)}{\mathbb{P}(Y \le k \mid x)} = e^{-\alpha_k} e^{x^\top \beta}.$$

- Raising x_j by one unit multiplies this by e^{β_j}, **the same at every k**.
- $\beta_j > 0$: larger x_j shifts mass to higher grades.

Treating nodules as independent, with $\eta_i = x_i^\top \beta$:

$$L(\alpha, \beta) = \prod_{i=1}^{n} \prod_{k=1}^{5} \Big[F(\alpha_k - \eta_i) - F(\alpha_{k-1} - \eta_i) \Big]^{1\{Y_i = k\}}$$

$$\ell(\alpha, \beta) = \sum_{i=1}^{n} \log \big[F(\alpha_{Y_i} - \eta_i) - F(\alpha_{Y_i - 1} - \eta_i) \big].$$

Score for β

Using $F' = F(1-F) =: f$ and writing $F_{ik} = F(\alpha_k - \eta_i)$, $f_{ik} = f(\alpha_k - \eta_i)$:

$$\frac{\partial \ell}{\partial \beta} = -\sum_i x_i \, \frac{f_{i,Y_i} - f_{i,Y_i - 1}}{F_{i,Y_i} - F_{i,Y_i - 1}}.$$

Maximisation

- $\nabla \ell = 0$ has no closed form.
- Solved iteratively by **BFGS** (quasi-Newton):

$$\theta^{(t+1)} = \theta^{(t)} + s_t B_t \nabla \ell(\theta^{(t)})$$

with step size s_t and $B_t \approx (-\nabla^2 \ell)^{-1}$.

- ℓ is concave in $(\alpha, \beta) \Rightarrow$ global maximum.

Interpretation and Prediction Rule

Odds ratios

Features are standardised, so $e^{\hat{\beta}_j}$ is the multiplicative change in the odds of a higher grade per **one standard deviation** of x_j, other predictors fixed.

- $e^{\beta_j} > 1$: higher grade more likely.
- $e^{\beta_j} < 1$: lower grade more likely.
- $e^{\beta_j} = 1$: no association.

Predicted grade

$$\hat{Y}_i = \underset{k}{\operatorname{argmax}} \Big[F(\hat{\alpha}_k - x_i^\top \hat{\beta}) - F(\hat{\alpha}_{k-1} - x_i^\top \hat{\beta}) \Big]$$

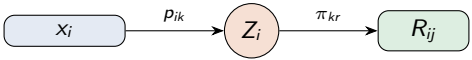
the *modal* grade of the fitted distribution.

Naive

The model takes the **median** as the true grade and ignores reader disagreement.

Why a Latent True Grade?

- Each nodule has up to 4 ratings R_{ij}, which frequently disagree.
- Median-taking discards information: a nodule rated $(3, 3, 3, 3)$ and one rated $(1, 2, 3, 5)$ both look like "3".
- The true class is unobserved; each observer is a noisy measuring device with an error-rate matrix.



structural model $\rightarrow$ latent grade $\rightarrow$ measurement model

Idea

The true grade $Z_i \in \{1, \ldots, 5\}$ is *unobserved*. Features determine Z_i through a proportional odds model, and the readers' ratings are noisy measurements of Z_i.

Latent-Grade Model Specification

Structural model (primary features x_i)

$p_{ik}(\alpha, \beta) = \mathbb{P}(Z_i = k \mid x_i) =$
$F(\alpha_k - x_i^\top \beta) - F(\alpha_{k-1} - x_i^\top \beta)$

Measurement model

$$\mathbb{P}(R_{ij} = r \mid Z_i = k) = \pi_{kr}, \qquad \sum_{r=1}^{5} \pi_{kr} = 1.$$

- One shared 5×5 matrix π for all readers.
- Ratings are conditionally independent given Z_i.

Parameters and data

- $\Psi = (\alpha, \beta, \pi)$.
- $C_{ir} = \#\{j : R_{ij} = r\}$: number of readers giving nodule i the rating r.
- Likelihood depends on ratings only through the counts C_{ir} (sufficient given Z_i).

Deriving the Observed-Data Likelihood

Setup for nodule i: true grade Z_i (unobserved), ratings $R_{i1}, \ldots, R_{im_i}$ from $m_i \le 4$ readers, and counts $C_{ir} = \#\{j : R_{ij} = r\}$.

1. **Joint probability of a grade and its ratings.** By the conditional independence of readers given Z_i:

$$\mathbb{P}(Z_i = k,\, R_i \mid x_i) = \underbrace{p_{ik}(\alpha, \beta)}_{\mathbb{P}(Z_i = k \mid x_i)} \prod_{j=1}^{m_i} \underbrace{\pi_{k, R_{ij}}}_{\mathbb{P}(R_{ij} \mid Z_i = k)} = p_{ik} \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

   (Grouping the m_i readers by the rating they gave turns the product over readers into a product over ratings r, with exponent C_{ir}.)

2. **Marginalise out the unobserved Z_i.** Only the ratings are observed, so sum over the five possible true grades (law of total probability):

$$\mathbb{P}(R_i \mid x_i) = \sum_{k=1}^{5} \mathbb{P}(Z_i = k,\, R_i \mid x_i) = \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

3. **Combine nodules.** Nodules are independent, so the likelihood is the product $\prod_i \mathbb{P}(R_i \mid x_i)$. Taking logs:

$$\ell(\Psi) = \sum_{i=1}^{n} \log \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

Observed vs. Complete-Data Likelihood

Observed (incomplete) data – the ratings only.

$$\ell(\Psi) = \sum_{i=1}^{n} \log \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

The sum *inside* the logarithm prevents the problem from separating, so direct maximisation is hard (and the surface need not be concave).

Complete data – if Z_i were observed, the log-likelihood would split into two simple pieces:

$$\ell_c(\Psi) = \sum_{i=1}^{n} \sum_{k=1}^{5} 1\{Z_i = k\} \Big[\underbrace{\log p_{ik}(\alpha, \beta)}_{\text{ordinal regression}} + \underbrace{\sum_{r=1}^{5} C_{ir} \log \pi_{kr}}_{\text{multinomial}} \Big].$$

EM strategy

Replace the unknown indicators $1\{Z_i = k\}$ by their conditional expectations, then maximise the resulting (simple) surrogate; repeat.

The EM Algorithm: E-Step

Given the current iterate $\Psi^{(t)}$, compute the **posterior probability of each true grade** using Bayes' rule:

$$w_{ik}^{(t)} = \mathbb{P}\big(Z_i = k \mid R_i, x_i, \Psi^{(t)}\big) = \frac{p_{ik}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{kr}^{(t)}\big)^{C_{ir}}}{\sum_{l=1}^{5} p_{il}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{lr}^{(t)}\big)^{C_{ir}}}.$$

Taking the conditional expectation of ℓ_c gives the Q-function:

$$Q(\Psi \mid \Psi^{(t)}) = \mathbb{E}\big[\ell_c(\Psi) \mid R, x, \Psi^{(t)}\big] = \sum_{i,k} w_{ik}^{(t)} \log p_{ik}(\alpha, \beta) + \sum_{i,k,r} w_{ik}^{(t)} C_{ir} \log \pi_{kr}.$$

The EM Algorithm: M-Step

$\Psi^{(t+1)} = \text{argmax}_\Psi\, Q(\Psi \mid \Psi^{(t)})$. The two terms of Q involve different parameters, so they are maximised separately.

Update of π (closed form)

Maximise $\sum_{i,r} w_{ik} C_{ir} \log \pi_{kr}$ subject to $\sum_r \pi_{kr} = 1$. Lagrangian:

$$\mathcal{L} = \sum_{i,r} w_{ik} C_{ir} \log \pi_{kr} - \lambda_k \Big(\sum_r \pi_{kr} - 1 \Big)$$

$\partial \mathcal{L} / \partial \pi_{kr} = 0 \Rightarrow \pi_{kr} \propto \sum_i w_{ik} C_{ir}$, so

$$\pi_{kr}^{(t+1)} = \frac{\sum_i w_{ik}^{(t)} C_{ir}}{\sum_i \sum_s w_{ik}^{(t)} C_{is}}.$$

Weighted relative frequency of rating r among nodules of grade k.

Update of (α, β)

Maximise $\sum_{i,k} w_{ik}^{(t)} \log p_{ik}(\alpha, \beta)$.

- No closed form.
- Computed with BFGS.

Algorithm

1. Initialise $\Psi^{(0)}$.
2. **E-step:** compute $w_{ik}^{(t)}$.
3. **M-step:** update $\pi^{(t+1)}$ and $(\alpha, \beta)^{(t+1)}$.
4. Stop when $|\ell(\Psi^{(t+1)}) - \ell(\Psi^{(t)})| < \epsilon$ (say, 10^{-6}); set $\hat{\Psi} = \Psi^{(t+1)}$.

Theorem (Wilks, 1938)

Under regularity conditions and if H_0 is true, as $n \to \infty$,

$$\Lambda = 2\Big[\sup_{\theta \in \Theta} \ell(\theta) - \sup_{\theta \in \Theta_0} \ell(\theta)\Big] \xrightarrow{d} \chi^2_r,$$

where r is the number of parameters fixed by H_0.

- Full model M_1 and restricted model $M_0 \subset M_1$: $\Lambda = 2(\hat{\ell}_1 - \hat{\ell}_0)$.
- Reject H_0 at level 0.05 when $\Lambda > \chi^2_{r,0.05}$ (e.g. 3.841 for $r = 1$).
- Both fits are *maximised*: setting $\beta_j = 0$ removes predictor j, and **all** other parameters are re-estimated in the restricted fit.
- Large-sample result; hypotheses here are composite, so it is used as an approximation.

Regularity Conditions for Wilks Test

1. **Interior point:** The true parameter $\theta_0 = (\psi_0, \lambda_0)^T$ lies in the interior of Θ, not on its boundary.
2. **Common support:** The support of $f(x \mid \theta)$, defined as $\{x : f(x \mid \theta) > 0\}$, is independent of θ.
3. **Smoothness:** The density $f(x \mid \theta)$ is three times continuously differentiable with respect to θ in a neighborhood $N(\theta_0)$.
4. **Information Matrix:** The expected Fisher information matrix, $I(\theta_0) = -\mathbb{E}_{\theta_0}\left[\nabla^2_\theta \log f(X \mid \theta_0)\right] = \mathbb{E}_{\theta_0}\left[(\nabla_\theta \log f(X \mid \theta_0))(\nabla_\theta \log f(X \mid \theta_0))^T\right]$, is finite and strictly positive definite.
5. **Bounded third derivatives:** There exists a function $M(x)$ such that $\mathbb{E}_{\theta_0}[M(X)] < \infty$ and $\left|\frac{\partial^3}{\partial\theta_j \partial\theta_l \partial\theta_m} \log f(x \mid \theta)\right| \leq M(x)$ for all $\theta \in N(\theta_0)$.

Tests Performed

Test	**Null hypothesis H_0**	r	**Reference**
Overall significance	$\beta = 0$	5 / 10	χ^2_5, χ^2_{10}
Drop-one (each predictor)	$\beta_j = 0$	1	χ^2_1
Semantic ratings jointly	five semantic β's $= 0$	5	χ^2_5
EM model: overall	$\beta = 0$	5	χ^2_5
EM model: drop-one	$\beta_j = 0$	1	χ^2_1
Prop. odds (one predictor)	$\beta_j^{(1)} = \cdots = \beta_j^{(4)}$	3	χ^2_3
Prop. odds (all predictors)	all equal across thresholds	15 / 30	χ^2_{15}, χ^2_{30}

- All tests are at level $\alpha = 0.05$, on the training set.
- Drop-one tests are *not* adjusted for multiple comparisons.

Profile-Likelihood Confidence Intervals

- For a single coefficient β_j, the **profile log-likelihood** maximises over all other parameters ψ:

$$\ell_p(b) = \max_{\theta \mid \beta_i = b} \ell(\theta).$$

  where $\theta = (\alpha, \beta)$.
- Invert the likelihood ratio test of $H_0 : \beta_j = b$. By Wilks with $r = 1$,

$$\mathrm{CI}_{95\%}(\beta_j) = \Big\{ b : \; 2\big[\ell(\hat{\theta}_{\mathrm{MLE}}) - \ell_p(b)\big] \le \chi^2_{1,0.05} = 3.841 \Big\}.$$

- Exponentiating the endpoints gives the interval for the odds ratio e^{β_j}.

Proportional Odds Assumption: Test

Restricted model M_0 (proportional odds): one coefficient vector β for all thresholds.
Full model M_1 (non-proportional): a separate coefficient at each threshold,

$$\text{logit}\,\mathbb{P}(Y_i \le k \mid x_i) = \alpha_k - x_i^\top \beta^{(k)}, \quad k = 1, \dots, 4.$$

Hypotheses (predictor j)

H_0 : $\beta_j^{(1)} = \beta_j^{(2)} = \beta_j^{(3)} = \beta_j^{(4)}$
H_1 : effect differs for at least one threshold.

- Likelihood ratio test : $\Lambda = 2(\hat{\ell}_{\text{non-PO}} - \hat{\ell}_{\text{PO}})$.
- Freeing one predictor adds 3 parameters: $r = 3$.
- Freeing all: $r = 15$ (primary), $r = 30$ (secondary).
- Applied to the **naive models**.

Why it matters

If rejected, a single odds ratio fails to summarize the data adequately.

Out-of-Sample Evaluation

- Models are evaluated **once** on the held-out test patients, and the coefficients are frozen.
- **Prediction:** the most likely grade. For EM, convert latent grades to rating probabilities via the measurement matrix:

$$\mathbb{P}(R = r \mid x_i) = \sum_{k=1}^{5} p_{ik}(\hat{\alpha}, \hat{\beta})\, \hat{\pi}_{kr}, \qquad \hat{Y}_i = \operatorname*{argmax}_r \mathbb{P}(R = r \mid x_i).$$

- **Error metrics** (against median grade Y_i):

$$\text{Misclass.} = \tfrac{1}{n} \sum_i 1\{\hat{Y}_i \neq Y_i\}, \quad \text{MAE} = \tfrac{1}{n} \sum_i |\hat{Y}_i - Y_i|, \quad \text{RMSE} = \Big(\tfrac{1}{n} \sum_i (\hat{Y}_i - Y_i)^2\Big)^{1/2}.$$

- **Baseline:** always predict grade 3 (most common in training).

Triage Rule

Rule

Flag nodule i for priority review if

$$\mathbb{P}(Y_i \geq 4 \mid x_i) > \tau.$$

On the training set, τ is set to the largest value that still catches 90% of the high-grade nodules. That τ is then applied unchanged to the test set. This keeps the test evaluation honest, since the cutoff wasn't tuned on test data.

Metrics

- **Sensitivity:** share of grade ≥ 4 nodules that are flagged
- **Workload:** share of all nodules flagged.
- **Precision:** share of flagged nodules with grade $\geq$4.

- For EM, $\mathbb{P}(Y \geq 4 \mid x)$ is taken on the *rating* scale: $\sum_{r \geq 4} \sum_k p_{ik} \hat{\pi}_{kr}$.
- The rule ranks nodules; it is not a diagnosis.

Primary Naive Model (Proportional Odds)

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
log_volume	1.306	0.064	3.693	(3.259, 4.193)
elongation	0.333	0.055	1.396	(1.254, 1.554)
flatness	−0.178	0.053	0.837	(0.754, 0.929)
solidity	−0.327	0.065	0.721	(0.635, 0.818)
compactness	0.048	0.059	1.049	(0.935, 1.177)

Table 2: $\hat{\alpha} = (-2.456, -1.068, 1.602, 3.291)$; max. log-likelihood $= -2193.908$

- **Size** dominates: one unit increase in log volume (about 4× volume) multiplies the odds of a higher grade by 3.69.
- Elongation and flatness are coded so that values near 1 mean not elongated and not flat. Elongation has OR 1.40, so less elongated nodules have higher odds of a higher grade. Flatness has OR 0.84, so flatter nodules have higher odds.

Secondary Naive Model: Adding Semantic Ratings

Predictor	$\hat{\beta}$	SE	OR	95% CI
log_volume	1.183	0.067	3.264	(2.865, 3.724)
elongation	0.321	0.055	1.378	(1.237, 1.537)
flatness	−0.075	0.055	0.928	(0.832, 1.034)
solidity	−0.158	0.069	0.854	(0.746, 0.977)
compactness	0.008	0.059	1.008	(0.898, 1.132)
spiculation	0.329	0.066	1.390	(1.221, 1.584)
lobulation	0.092	0.065	1.097	(0.965, 1.247)
margin	−0.279	0.074	0.757	(0.655, 0.874)
texture	0.067	0.067	1.070	(0.939, 1.219)
sphericity	0.014	0.050	1.014	(0.919, 1.120)

Table 3: $\hat{\alpha} = (-2.518, -1.098, 1.661, 3.404)$; log-lik. $= -2150.257$

- Spiculation OR 1.39; sharper margin OR 0.76 ; Size remains largest (OR 3.26).

Latent-Grade (EM) Estimates

Predictor	$\hat{\beta}$ (EM)	OR (EM)	OR (Naive)
`log_volume`	1.682	5.375	3.693
`elongation`	0.471	1.601	1.396
`flatness`	−0.323	0.724	0.837
`solidity`	−0.583	0.558	0.721
`compactness`	0.010	1.010	1.049

Table 4: $\hat{\alpha} = (-2.983, -0.494, 1.968, 4.024)$

- All six starting values converge to the same log-likelihood, −5190.172: no sign of local maxima.
- Same directions as naive; every main OR is **further from** 1 under EM.
- Plausible reason: modelling reader noise reduces attenuation.

Estimated Rating Probabilities $\hat{\pi}_{kr}$

Latent	R=1	R=2	R=3	R=4	R=5
1	**0.956**	0.044	0.000	0.000	0.000
2	0.103	0.395	**0.499**	0.002	0.000
3	0.035	0.255	**0.541**	0.169	0.000
4	0.000	0.092	0.313	**0.397**	0.197
5	0.004	0.000	0.099	0.194	**0.703**

Table 5: Row k: distribution of a reader's rating given true grade k

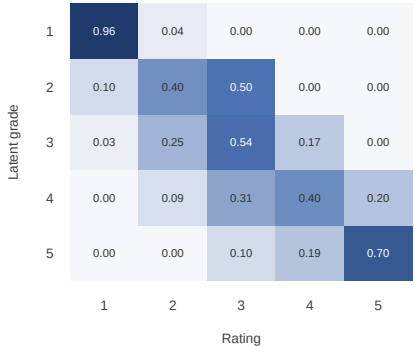


Figure 2: Heatmap of $\hat{\pi}$

Latent grades $1, 3, 4, 5$ are most often rated as themselves; latent grade 2 is rated 3 more often than 2.

Overall and Joint Significance

Model	Comparison	$\hat{\ell}_1$	$\hat{\ell}_0$	Λ	p-value
Naive primary	$\beta = 0$ ($r = 5$)	-2193.908	-2578.524	769.231	< 0.001
Naive secondary	$\beta = 0$ ($r = 10$)	-2150.257	-2578.524	856.534	< 0.001
Semantic ratings	primary vs. secondary ($r = 5$)	-2150.257	-2193.908	87.303	< 0.001
EM	$\beta = 0$ ($r = 5$)	-5190.172	-5606.396	832.449	< 0.001

Table 6: Likelihood ratio tests (training set)

- Predictors are jointly associated with malignancy grade in every model.
- Semantic ratings significantly improve the model.
- EM and naive log-likelihoods use different data (individual ratings vs. median) and are only compared within each family.

Drop-One Tests: Naive Models

	Primary			Secondary		
Predictor	$\hat{\ell}_0$	Λ	p	$\hat{\ell}_0$	Λ	p
`log_volume`	-2429.043	470.270	$< .001$	-2320.867	341.220	$< .001$
`elongation`	-2212.687	37.556	$< .001$	-2167.237	33.961	$< .001$
`flatness`	-2199.565	11.314	$< .001$	-2151.176	1.838	0.175
`solidity`	-2206.846	25.874	$< .001$	-2152.895	5.276	0.022
`compactness`	-2194.241	0.665	0.415	-2150.266	0.018	0.893
`spiculation`	–	–	–	-2162.799	25.084	$< .001$
`lobulation`	–	–	–	-2151.258	2.002	0.157
`margin`	–	–	–	-2157.508	14.501	$< .001$
`texture`	–	–	–	-2150.770	1.026	0.311
`sphericity`	–	–	–	-2150.297	0.080	0.777

Table 7: Drop-one LRT, $r = 1$

Primary: all but compactness significant. Secondary: log volume, elongation, solidity, spiculation, margin significant; flatness no longer is.

Drop-One Tests: EM Model

Predictor	$\hat{\ell}_0$	Λ	p-value
`log_volume`	-5424.701	469.058	< 0.001
`elongation`	-5210.130	39.917	< 0.001
`flatness`	-5200.096	19.848	< 0.001
`solidity`	-5212.612	44.880	< 0.001
`compactness`	-5190.181	0.018	0.894

Table 8: Full model $\hat{\ell}_1 = -5190.172$

- Same four features significant as in the primary naive model; compactness is not.
- Log volume again has by far the largest statistic.
- Each restricted model is refitted by EM.

Proportional Odds Assumption: Rejected

	Primary		Secondary	
Predictor	Λ	p-value	Λ	p-value
`log_volume`	136.258	< 0.001	137.293	< 0.001
`elongation`	7.979	0.046	9.285	0.026
`flatness`	48.578	< 0.001	53.972	< 0.001
`solidity`	26.207	< 0.001	17.305	< 0.001
`compactness`	46.604	< 0.001	40.818	< 0.001
`spiculation`	–	–	4.703	0.195
`lobulation`	–	–	12.084	0.007
`margin`	–	–	76.420	< 0.001
`texture`	–	–	61.760	< 0.001
`sphericity`	–	–	6.303	0.098
All predictors	**206.089**	< 0.001	**290.227**	< 0.001

Table 9: LRT of proportional odds ($r = 3$ per predictor; $r = 15$ / 30 for all)

Every primary feature violates the assumption; in the secondary model only spiculation and sphericity do not.

Test-Set Prediction Errors

	Error			Reduction vs. baseline (%)		
Model	Misclass.	MAE	RMSE	Misclass.	MAE	RMSE
Baseline	0.512	0.676	1.002	–	–	–
Naive Primary	0.507	0.652	0.974	0.95	3.58	2.81
Naive Secondary	**0.486**	**0.616**	**0.940**	4.96	8.94	6.21
EM	0.510	0.688	1.031	0.24	−1.79	−2.85

Table 10: Held-out patients (828 nodules)

- Naive secondary is best, but only modestly better than always predicting grade 3.
- Naive primary improves only slightly; EM does *not* improve on the baseline in MAE or RMSE.
- Exact-grade prediction is poor for all models.

Predicted Grade Distribution on the Test Set

	1	2	3	4	5
Actual median	89	173	404	114	47
Naive Primary	44	20	676	46	41
Naive Secondary	44	29	659	57	38
EM	65	0	702	0	60

Table 11: Nodules predicted at each grade

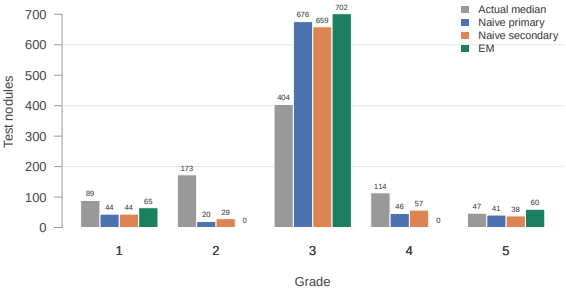


Figure 3: Actual vs. predicted counts

All models over-predict grade 3; the EM model never predicts grades 2 or 4, because latent grades 2 and 4 are mostly rated 3.

Triage Rule Performance

Model	τ	Sens.	Work.	Prec.	Missed
Naive Primary	0.139	0.913	0.470	0.378	14
Naive Secondary	0.146	0.894	0.426	0.409	17
EM	0.147	0.907	0.441	0.400	15

Table 12: Test set: 161 of 827 nodules (19.5%) have grade ≥ 4

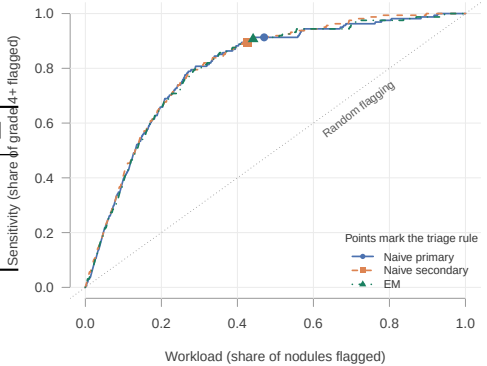


Figure 4: Sensitivity vs. workload

The three curves nearly coincide: the models rank nodules very similarly, and all are well above random flagging.

Policy Recommendation and Limitations

Recommendation

Use the **naive primary model** as the starting point for a triage tool:

- Reviewing flagged nodules first, a radiologist sees 91.3% of high-grade nodules within the first 47.0% of cases.
- About 38% of flagged nodules are high grade.
- Simplest model; needs no semantic ratings; the differences between models are too small to justify added complexity.

Caution

The rule must **never** be used to dismiss a nodule. It misses 14 of the 161 high-grade test nodules, which should still receive routine review.

Limitations

- **Suspicion, not diagnosis:** LIDC-IDRI malignancy is radiologist suspicion, not pathology-confirmed; TCIA diagnosis data for a small subset could enable external validation.
- **Dependence on drawn outlines:** geometric features come from radiologist outlines; a deployed tool would need automatic segmentation (not assessed).
- **Generalisation:** a research dataset from few US centres; use in other settings (e.g. Indian public hospitals) is untested.

Conclusions

1. **Size dominates:** one unit of log volume multiplies the odds of a higher grade by 3.69. Log volume, elongation, flatness and solidity are significant; compactness is not.
2. **Proportional odds is rejected** for both naive models: size effects grow across thresholds; flatness and compactness change direction.
3. **Exact-grade prediction is poor:** secondary model 48.6% misclassification vs. 51.2% baseline; EM is worse. Average reader disagreement is 0.74 grades.
4. **Ranking is useful:** the primary triage rule captures 91.3% of high-grade nodules while flagging 47.0% of all nodules.

Thank you!

Questions?

